

3.8.1 Centre Assessed

Route T (PSA/ISA)

Each centre assessed unit comprises

- Practical Skills Assessment (PSA)
- Investigative Skills Assignment (ISA).

The PSA consists of the centre's assessment of the candidate's ability at the end of the course to demonstrate practical skills; thus, candidates should be encouraged to carry out practical and investigative work throughout the course. This work should cover the skills and knowledge of How Science Works (Section 3.7) and in Sections 3.3 and 3.6.

The ISA has three stages where candidates

- undertake practical work and collect data
- process the data
- complete a written ISA test.

Each stage must be carried out under controlled conditions but may be scheduled at a time convenient to the centre. The written test must be completed in a single, uninterrupted session.

The ISA is set externally by AQA, but internally marked, with marking guidelines provided by AQA.

In a given academic year two ISAs at each of AS and A2 will be provided.

and conduct of practical work, and the manner in which equipment is used.

The assessment criteria

In the context of material specified in the relevant AS or A2 specification, candidates will be assessed in the following skills

- following instructions
- selecting and using equipment
- organisation and safety

Descriptors for these three skills areas are provided for 0, 1 and 2 marks.

Candidates should be awarded marks which reflect their level of performance at the end of the course.

AQA may wish to ask for further supporting evidence from centres in relation to the marks awarded for the PSA. Centres should therefore keep records of their candidates' performances in their practical activities throughout the course. (For example, a laboratory diary, log or tick sheet.)

Further guidance for the awarding of marks for the PSA will be provided in the Teacher Resource Bank.

Use of ICT during the PSA

Candidates are encouraged to use ICT where appropriate in the course of developing practical skills, for example in collecting and analysing data.

The Assessment Criteria

Following instructions	Selecting and using equipment	Organisation and safety
0 marks Able to follow instructions involving standard procedures only when guidance is given.	0 marks Able to select appropriate laboratory equipment only when guidance is given. Able to make measurements with assistance.	0 marks Works in a disorganised manner. Works safely only with constant supervision.
1 mark Able to follow instructions involving standard procedures but unable to follow instructions involving complex procedures without further guidance.	1 mark Able to select appropriate laboratory equipment. Able to make accurate measurements but requires guidance on making repeat measurements.	1 mark Works in an organised manner. Works with due regard to safety, but needs occasional guidance or reminders.
2 marks Able to follow instructions involving both standard and complex procedures without guidance.	2 marks Able to select appropriate laboratory equipment. Able to make accurate measurements and to recognise when it is appropriate to repeat measurements.	2 marks Works in an organised manner. Works with due regard to safety, showing competence in risk management.

Investigative Skills Assignment (ISA)

The Investigative Skills Assignment carries 44 marks and has three stages.

Stage 1: Collection of data

Candidates carry out practical work following an AQA task sheet. Centres may use the task sheet as described or may make minor suitable modifications to materials or equipment, following AQA guidelines. Details of any amendments made to the task sheet must be agreed in writing with the AQA Assessment Adviser. The task may be conducted in a normal timetabled lesson but must be under controlled conditions.

For AS, candidates collect raw data and represent it in a table of their own design or make observations that are recorded on the *Candidate Result Sheet*. The candidates' work must be handed to the teacher at the end of the session. The teacher assesses the candidates' work following AQA marking guidelines.

For A2, candidates collect raw data on the *Candidate Result Sheet*. The candidates' work must be handed to the teacher at the end of each session. The raw data is not assessed by the teacher.

There is no specified time limit for this stage.

Stage 2: Processing of data

The teacher returns the candidates' data from Stage 1 (on the *Candidate Result Sheet*).

For AS, the teacher instructs the candidates to process the data (e.g. calculate means or rates of

reaction) and plot an appropriate graph. The teacher must not instruct the candidates on the presentation of the data or on the choice of graph or chart.

For A2, the teacher instructs the candidates to process the data and carry out a suitable statistical test. The teacher must not instruct the candidates on the choice, implementation and interpretation of the statistical test.

For both AS and A2, stage 2 may be done in normal lesson time and must be done in a single session under controlled conditions. Both the raw and the processed data must be handed to the teacher at the end of the session. The teacher assesses the candidates' work to AQA marking guidelines.

Stage 1 and Stage 2 may be done in the same session. There is no specified time limit for Stage 2.

Stage 3: The ISA written test

The ISA written test should be taken as soon as convenient after completion of Stage 2, and under controlled conditions. Each candidate is provided with an ISA test and their completed material from Stages 1 and 2. The teacher uses the AQA marking guidelines to assess the ISA test.

The ISA test is in two Sections.

Section A

This consists of a number of questions relating to the candidate's own data.

Section B

At the start of this section, candidates are supplied with additional data on a related topic. A number of

questions relating to analysis and evaluation of the data then follow.

The number of marks allocated to each section may vary slightly with each ISA test.

Use of ICT during the ISA

ICT may be used during the ISA but teachers should note any restrictions in the ISA marking guidelines or Teachers' Notes. Use of the internet is not permitted.

Candidates absent for the practical work

A candidate absent for the practical work should be given an opportunity to carry out the practical work before they sit the ISA test. This may be with another group or at a different time. In exceptional cases when such arrangements are not possible, the teacher can supply a candidate with class data. In this case the candidate cannot be awarded marks for Stage 1, but may be awarded marks for Stage 2.

Material from AQA

For each ISA, AQA will provide:

- Teachers' Notes
- Task sheet
- ISA test
- Marking guidelines.

This material must be kept under secure conditions within the centre. If it is to be used in more than one session, then the centre must ensure security of the material between sessions. Further details regarding this material will be provided.

Security of assignments

Completed, marked ISAs should be treated like examination papers and kept under secure conditions until the publication of results.

General Information

Route T

Administration

In any year a candidate may attempt either or both of the two ISAs.

For each candidate, the teacher should submit a total mark to AQA comprising:

- the PSA mark
- the better ISA mark (if two have been attempted).

The ISA component of this mark must come from one ISA only, i.e. the marks awarded for individual stages of different ISAs cannot be combined.

Candidates may make only one attempt at a particular ISA. Redrafting is **not** permitted at any stage during the ISA.

The mark must be submitted by the due date in the academic year for which the ISA was published.

Only Internally Assessed Units from the Biology specification can contribute towards an AS or A Level Award in Biology. **Human Biology Units cannot be used towards a Biology qualification.**

Work to be submitted

For each candidate in the sample the following materials must be submitted to the moderator by the deadline issued by AQA.

- the candidate's data from Stage 1 and 2 (on the *Candidate Result Sheet*)
- the ISA written test, and
- a *Candidate Record Form* showing the marks for the ISA and the PSA.

In addition each centre must provide

- *Centre Declaration Sheet*
- Details of any agreed amendments to the task sheet with information supporting the changes from the AQA Assessment Adviser.

Working in groups

For the PSA, candidates may work in groups provided that any skills being assessed are the work of individual candidates. For the ISA further guidance will be provided in the Teachers' Notes.

Other information

Section 6 outlines further guidance on the supervision and authentication of internally assessed units.

Section 6 also provides information in relation to the internal standardisation of marking for these units. Please note that the marking of both the PSA and the ISA must be internally standardised as stated in Section 6.4.

Further support

AQA supports the units in a number of ways.

- AQA holds annual standardising meetings on a regional basis for all internally assessed components. Section 6 of this specification provides further details about these meetings
- A Teacher Resource Bank which includes further information and guidance
- Assessment Advisers are appointed by AQA to provide advice on internally assessed units. Every centre is allocated an Assessment Adviser.

The Assessment Advisers can provide guidance on

- issues relating to the carrying out of tasks for assessment
- application of marking guidelines.

Any amendments to the ISA task sheet must be discussed with the Assessment Adviser and confirmation of the amendments made must be submitted to the AQA moderator.